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System Identification of Robotic Fish Swimming Reveals Roles of Reactive and Resistive Fluid Force in Two Distinct Swimming Gaits

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ABSTRACT

Understanding how fish generate thrust across diverse swimming gaits is central to bio-inspired robotics. Here, we combine particle image velocimetry (PIV), system identification, and simulation to uncover the hydrodynamic mechanisms underlying swimming of a modular robotic fish (μ Bot). By modulating motor frequency, μ Bot exhibited two distinct gaits: a low–medium frequency traveling-wave “fish-like” gait shedding two pairs of vortices per cycle, and a high-frequency standing-wave “resonant” gait shedding double rows of single vortices of opposite sign per cycle. PIV revealed that thrust in the resonant gait is dominated by tail pressure (reactive), whereas such pressure contributions are negligible in the traveling-wave gait with thrust primarily generated by drag forces (resistive). We developed an analytical model incorporating reactive and resistive force components and identified its parameters using genetic algorithms to fit experimental data. The fitted model reproduced swimming speed and body kinematics with high accuracy and revealed gait-dependent variations in hydrodynamic parameters. Sensitivity analyses confirmed that only a subset of force components influenced swimming performance. Finally, energetic analyses showed resonance behavior in standing-wave gait but not in traveling-wave gait. This integrative framework highlights the distinct roles of reactive versus resistive forces across gaits, advancing the design and control of efficient fish-inspired robots.

1 | Introduction

Fish species have evolved with astonishingly diverse forms and locomotion capabilities in various underwater environments [1–4]. In particular, body and/or caudal-fin (BCF) swimming, which relies primarily on undulatory motions in body and caudal-fin for thrust generation, has served as a major design template for novel underwater vehicles, with the potential for improved efficiency and maneuverability than those using conventional propellers [5–8]. BCF swimmers use an array of muscles or actuators to deform their body and caudal-fin, enabling interactions with the surrounding fluid and the generation of hydrodynamic forces. The interplay between hydrodynamic forces and muscle-

generated torques shapes their swimming gaits (i.e., body and caudal-fin deformation) and determines overall swimming performance. This represents a controlled fluid–structure interaction (FSI) process that is inherently high dimensional and nonlinear.

Recent decades have seen major advances in understanding this process using experiments on living fish [9, 10] or robotic fish [11, 12] with flow and gait measurements and computational fluid dynamics (CFD) simulation [13–15]. Using particle image velocimetry (PIV), two main types of wake structures have been identified in fish BCF swimming [16]. The first pattern is reversed von Karman vortex (i.e., the fish sheds double row of single vortices of opposite sign per cycle; denoted as “2S” pattern) that

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correspond to a thrust wake, which has been observed in fish species such as zebrafish [17], mullet [18], and trout [19]. The second pattern resembles a double-row of double vortices (i.e., the fish sheds two pairs of vortices per cycle; denoted as “2P” pattern). This wake pattern is observed in eel [20], larval zebrafish [21], and Kuhli loach [22]. Using computational fluid dynamics (CFD) simulations and fish-inspired robot experiments, these patterns have not only been replicated but also expanded upon to include novel variants. For example, “2S” wake pattern was observed in CFD simulation of eel [23] and tuna [24]; Panah and Buchholz produced “2S” wake structure in the wake of a plunging foil [25]. With CFD simulation of foldable structure, Daghooghi and Borazjani produced “2P” wake pattern which changes along with the structure’s kinematics [26]; Using a compliant fish-inspired robot, Epps et al. observed “2P” wake pattern which varies with the undulation frequency of caudal-fin [27]; Additionally, a hybrid pattern combining elements of “2S” and “2P” (designated as “2S + 2P”) has emerged from further CFD studies [28, 29]. Despite these knowledge, the precise mechanisms for generating these wake patterns and their relation to the kinematics of the swimmers remain unclear.

Although experiments with living or robotic fish enable some direct measurements of swimming performance and the underlying physical processes [30], they remain limited by various experimental and technological constraints. For example, it is nearly infeasible to directly measure hydrodynamic forces using pressure sensors [31–33]. Consequently, it remains challenging to reveal the complete swimming mechanisms with limited experimental data, such as PIV results alone. On the other hand, CFD simulation solves Navier–Stokes equations based on prescribed swimming kinematics or in limited condition the complete entire FSI process in simplified forms [34, 35]. CFD usually provides detailed information regarding the pressure, force, and flow field and therefore has been the primary methodology in recent decades to study fish swimming. However, CFD remains computationally costly, and it is nearly impractical to apply CFD for robot design, optimization, or controller design [36–38].

Alternatively, analytical models of fish swimming [39–41] that are computationally-efficient, can be used to predict the hydrodynamic forces acting on the bodies of the swimmers, and have served as the primary approach to understand of fish swimming prior to the prevalence of experimental and computational approaches. The most widely used modeling method for fish swimming is based on the large amplitude elongated body theory (LAEBT) or Lighthill model [42]. In the LAEBT, the 3D swimming is reduced to a 2D model using potential flow theory that calculates the reactive force (pressure and added mass force) based on the fluid in a control volume attached to the fish body. Compared with the CFD solution, LAEBT can yield acceptable accuracy in limited conditions (i.e., when calibrate for a small number of swimming gaits), but demands significantly lower computational costs [43]. In recent decades, there are also several attempts to improve the Lighthill model and adapt it to a mobile multi-link system for robotics [44, 45].

The main challenges in developing an accurate analytical model for fish-inspired robotics center on the estimations of model parameters associated with various hydrodynamic mechanisms used to construct the model. These hydrodynamic mechanisms, including pressure forces, added mass forces, and circulatory based forces (e.g., friction, drag, lift), only model one aspect of force production in separate simplified conditions (e.g., assuming

potential flow). While these mechanisms can be coupled with each other, they are often combined as independent terms in the mathematical model to estimate the total force calculation, usually with empirical estimation of model parameter or with idealized values [46,47]. As a result, these analytical models often have limited local validity, and it is unclear how the hydrodynamic parameters could change with swimming gaits, morphologies, and fluid properties (e.g., Reynolds number).

To date, there are three methods applied to estimate the hydrodynamic parameters in analytical models. The first method is based on standard theoretical models [48–50], hydrodynamic parameters of which were calculated in idealized fluid conditions. While this method can yield critical insights in the hydrodynamic mechanisms, it is often not suitable for robot designs with complex morphologies (e.g., irregular shapes) and gaits (e.g., large amplitude, spatiotemporally varying body and fin deformation). The second method is to use empirical estimation based on direct force or PIV measurement from fluid experiments [51–56]. This method can yield accurate parameter estimations local to the experimental conditions, but it is limited to drag and friction coefficients and remain challenging or less accurate for added mass coefficient and pressure forces coefficient. The third method is to use system identification methods based on motion data from experiments. As the robot motion data can be captured more easily by external cameras, it is experimentally more practical for model parameter estimation [57, 58–61]. For example, Raj and Thakur directly implemented same prescribed gait for simulation and experiment and estimated the hydrodynamic parameters by matching the experimental and simulated swimming speed [61]. However, there can be redundancies among the hydrodynamic mechanisms or parameters in predicting the swimming speed, and the coupled effects among those parameters were often not modeled explicitly. These have rendered the identification process more challenging to yield a physically meaningful solution, which may not be found solely by data-driven system identification process. Furthermore, authors’ previous work [62] shows that swimming speed is sensitive to only a small number of model parameters for certain swimming gaits, so merely matching the speed without accounting for swimming gaits makes it infeasible to identify all relevant model parameters.

To address the above challenges, we developed a novel system identification approach by combining experiments on a robotic swimmer (named μ Bot), PIV experiments for μ Bot wake visualization, and computationally-efficient numerical simulation based on analytical models of μ Bot swimming. In authors’ previous work, μ Bot, a magnetic, modular, undulatory swimming robot, was developed as robophysical model to understand fish-inspired BCF swimming [44, 62, 63]. The μ Bots have variable number of electromagnetic actuators, which can apply spatially distributed torque on the μ Bot body structure. The actuators receive motor control signals from Central Pattern Generators (CPGs), which can be optimized via reinforcement learning in experiments for maximizing a performance measure, e.g., forward swimming speed [62] and turning angle [64]. It has been shown that the swimming gaits of μ Bot are highly sensitive to the frequency of motor control. In particular, two types of swimming gaits, i.e., Standing Wave “Resonant” Gaits (SW “Res” Gaits) under high frequency and Traveling Wave “Fish-Like” Gaits (TW “FL” Gaits) under low frequency, emerged as a result of motor frequency modulation.

In this work, we aim to identify the hydrodynamic model parameters of μ Bot in steady forward swimming by leveraging experimentally measured swimming kinematics and flow visualization. Based on the identified model we further (1) examined the sensitivities of swimming gaits and speed to identified hydrodynamic parameters, (2) examined roles of various hydrodynamic mechanisms in thrust generation and identified the force balance that led to steady forward swimming, and (3) analyzed the energetics of μ Bots, including those of the actuators (i.e., work loops) and the degree of resonance involved in the gait generation.

The system identification process is summarized in Figure 1. We first developed an analytical model of μ Bot-4 (μ Bot with four actuators) that includes (1) a hydrodynamic model with four types of forces (pressure force, added mass force, drag, and friction); (2) rigid body dynamics of robot head, body, peduncle, and caudal-fin segments; and (3) electromagnetic actuator model. The analytical model was then simulated using Gazebo with Open Dynamic Engine (ODE), which represented the motor-controlled FSI process of μ Bot and predicted both the swimming gaits and speed. We applied identical motor control signals to both the physical μ Bot in experiments and the simulated μ Bot in Gazebo and used a Genetic Algorithm (GA) to iteratively

update the model parameters to match the measured swimming gaits and speed with those produced by the model-based simulation. In this process, we reduced model redundancies by incorporating assumptions informed by PIV measurements of the μ Bot's wake structure and pressure fields, which enabled the identification of a unique set of model parameters.

The remainder of the study is organized as follows. Section 2 presents the mathematical modeling of μ Bot in terms of hydrodynamic model, actuator model, and rigid body model. Section 3 describes the experiment setup for the PIV techniques. Section 4 describes the system identification procedure and validation of modeling parameters. Section 5 presents the analyses of the vorticity fields and pressure fields and applies the identified model to reveal the roles of different hydrodynamic mechanism in thrust and gait generations. Finally, section 6 summarizes the main findings and provides concluding remarks.

2 | Mathematical Modeling of μ Bot

To simulate μ Bot swimming, a mathematical model was developed to calculate the torque generated by motor and the hydrodynamic

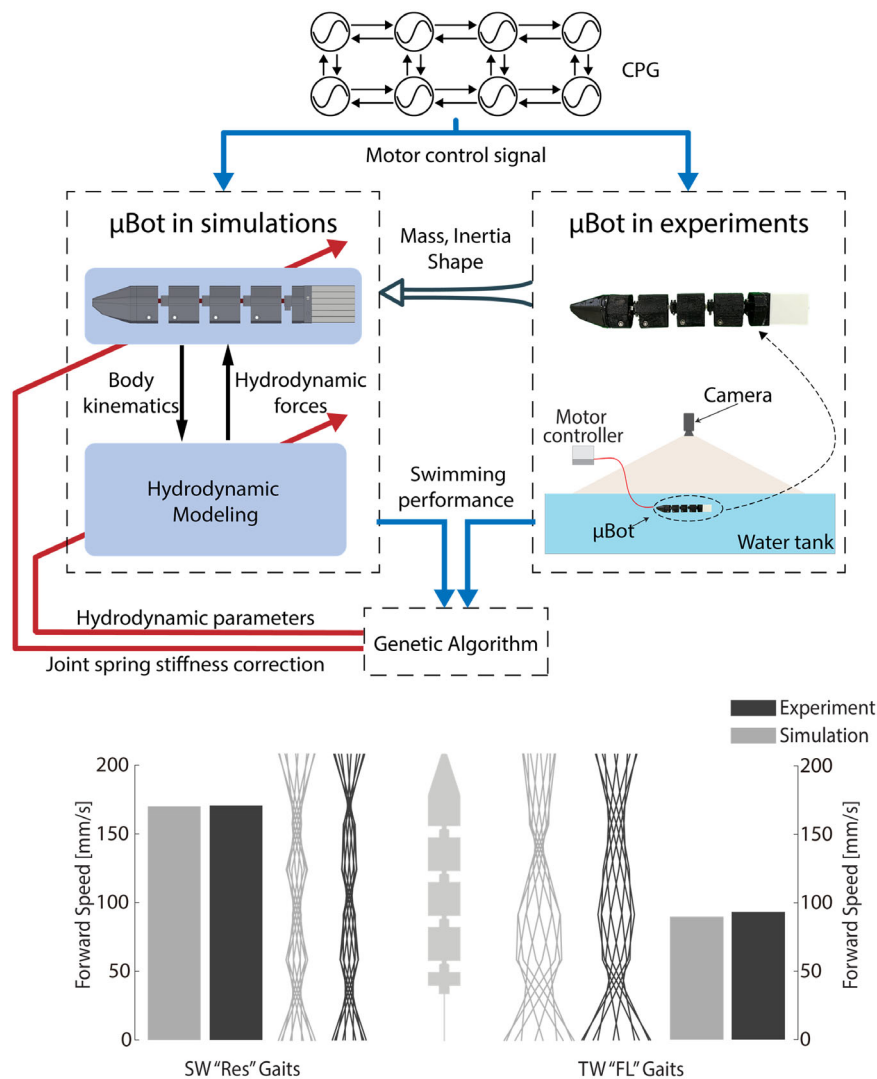


FIGURE 1 | Overview of the system identification method. We implemented the motor control signal from experiments into simulations, updated the modeling parameters to match the swimming performance predicted from simulation with data from experiments.

force and torque acting on μ Bot head, body, and caudal-fin segments (shown in Figure S1c). The hydrodynamic model was based on the reactive theory to estimate pressure force and added mass forces [65], and the resistive theory to estimate drag and friction forces [66]. The modeling approach is adapted and revised based on those used by our previous work [44] and Mathieu et al. [45]. To model the μ Bot-4 for subsequent simulations, several assumptions are made. First, the mass and moment of inertia of each rubber suit are lumped into those of the anterior segment where the rubber suit is mounted, and the rubber suit is simplified as a torsional spring at each joint. Second, the caudal-fin is simplified as a rigid plate connected to the peduncle segment via a torsion spring. Note that, although the caudal-fin can be flexible, it remains nearly flat during the experiments, and only has a small curvature during the experiment [44]. Third, as the μ Bot-4 swims on a horizontal plane during the experiments, only the planar motion and dynamics are modeled and calculated in the simulations.

2.1 | Overview of μ Bot Design and Motor Control

The μ Bots used modular designs with modifiable morphology, ease of assembly, and relatively small size compared with other robotic fish designs [67], facilitating rapid prototyping with diverse design parameters such as body shape, size, and length; its small size also allows for easy experiments in lab conditions. A more detailed description of μ Bot and its capabilities can be found in our previous work [61, 63, 68].

In this work, μ Bot-4, with four actuators and Degrees-of-Freedoms (DoF), is composed of a head segment, three body segments, a peduncle segment, and a caudal-fin (Figure 2a). A layer of rubber suit covers the gap between segments, provides stiffness between segments, and seals the robot internal hardware from water, while also expanding the robot hydrodynamic surface area. Both head and body modules include custom-designed electromagnetic actuators for generating body bending torque. A caudal-fin, designed as a thin layer of silicone rubber reinforced by carbon fiber rods, is mounted on the peduncle segment. A fully assembled μ Bot-4 measures 25 mm in width, 35 mm in depth, and 170 mm in length. The robot density is tuned so that it achieves slightly positive buoyancy.

μ Bot-4 was torque-controlled by electromagnetic motors. The swimming gaits/kinematics of μ Bot-4 are not prescribed; instead, they emerge from the FSI via motor control (voltage inputs to control actuators' torque). In our previous work [62], the motor control signal was generated by a central pattern generator (CPG), whose parameters were optimized to maximize the swimming speed with a Reinforcement Learning (RL). In this work, we used the RL-learned motor control signals (Figure 2c) from our previous work for our system identification and varied its frequency to study the frequency modulation of wake structures and thrust generation mechanisms in PIV experiments (Figure 2b).

2.2 | Actuator Model

For the electromagnetic motors on μ Bot, each of them has a coil mounted on a rotating coil arm around a pivot joint, while the coil is placed in between two permanent magnets pointed closely to each other with identical polarity, therefore opposing each other, (shown in Figure S1d). With voltage applied, the coil generates

lateral forces, which in turn creates actuation torque via the rotating arm that rotates the subsequent segment, and the torque can be reversed by simply reversing the applied voltage.

Based on the work of Alessio and coworkers [69] and our previous work [44], the model of electromagnetic can be simply expressed as:

$$T_{act,j} = \frac{Vol_j - k_{EMF} \dot{\theta}_j}{R} \cdot k_T \quad (1)$$

where T_{act} is torque generated by actuator, Vol_j is applied voltage, R is the resistivity of coil, k_T is actuator torque constant, and k_{EMF} is back EMF constant. This model was calibrated by torque measured experimentally at zero rotation position. The model shows that both the actuator torque constant (k_T) and back EMF constant (k_{EMF}) remain nearly constant across the usable range of coil rotation, which are calculated as 1.26Nmm/A for k_T , and 1.26mV · s for k_{EMF} .

2.3 | Reactive Force Model

Reactive force can be modeled by assuming the fluid is incompressible and irrotational. Reactive force model contains two components, added mass force ($\mathbf{f}_{add}$) and torque ($\mathbf{c}_{add}$), and pressure force ($\mathbf{f}_{pre,j}^-$ and $\mathbf{f}_{pre,j}^+$) (shown in Figure S1b). Applying Lighthill's model, a control volume, termed as added-mass volume (AMV) and attached to each segment, can be defined with two boundaries $\Pi_j(0)$, $\Pi_j(l_j)$. The change of linear momentum and angular momentum of water within the AMV are the added mass force and torque, which can be expressed as following:

$$\begin{bmatrix} \mathbf{f}_{add} \\ \mathbf{c}_{add} \end{bmatrix} = C_a \cdot \begin{bmatrix} -\bar{m}_j l_j \omega_j v_j + \frac{1}{2} \bar{m}_j l_j^2 \omega_j^2 \\ -\bar{m}_j l_j a_j - \frac{1}{2} \bar{m}_j l_j^2 \dot{\omega}_j \\ 0 \\ 0 \\ 0 \\ -\left[\frac{1}{2} \bar{m}_j l_j^2 (a_j - \omega_j u_j) + \frac{1}{3} \bar{m}_j l_j^3 \dot{\omega}_j + \bar{m}_j l_j u_j v_j \right] \end{bmatrix} \quad (2)$$

where l_j is segment length, ω_j is angular velocity, v_j is linear velocity of $\Pi_j(0)$ projected onto $\hat{y}_j$, a_j is linear acceleration of $\Pi_j(0)$ projected onto $\hat{y}_j$, and v_j' is linear velocity of boundary $\Pi_j(l_j)$ projected on $\hat{y}_j$ while the coordinate frame is shown in Figure S1a. Details can be found in our previous work [44].

In LAEBT, fluid outside AMV is assumed to have no velocity so that $\mathbf{f}_{pre,j}$ can be calculated simply with velocity of fluid within boundaries ($\Pi_j(0)$, $\Pi_j(l_j)$) via unsteady Bernoulli's principle. However, this assumption with abrupt change of velocity from inside to outside of the control volume can potentially lead to overestimation of the pressure force. Therefore, we introduced a correction coefficient C_p for pressure force. Thus, the pressure forces take the form:

$$\mathbf{f}_{pre,j}(\mathbf{x}) = C_a C_p \begin{bmatrix} \frac{1}{2} \cdot \bar{m}(\mathbf{x}) \cdot v_j^2(\mathbf{x}) \\ 0 \\ 0 \end{bmatrix} \quad (3)$$

The final expression of reactive force can be expressed as a function of linear velocities and angular velocity of each segment, dimensions of segment and two empirical coefficients C_a , C_p , while the reactive force vector can be written as

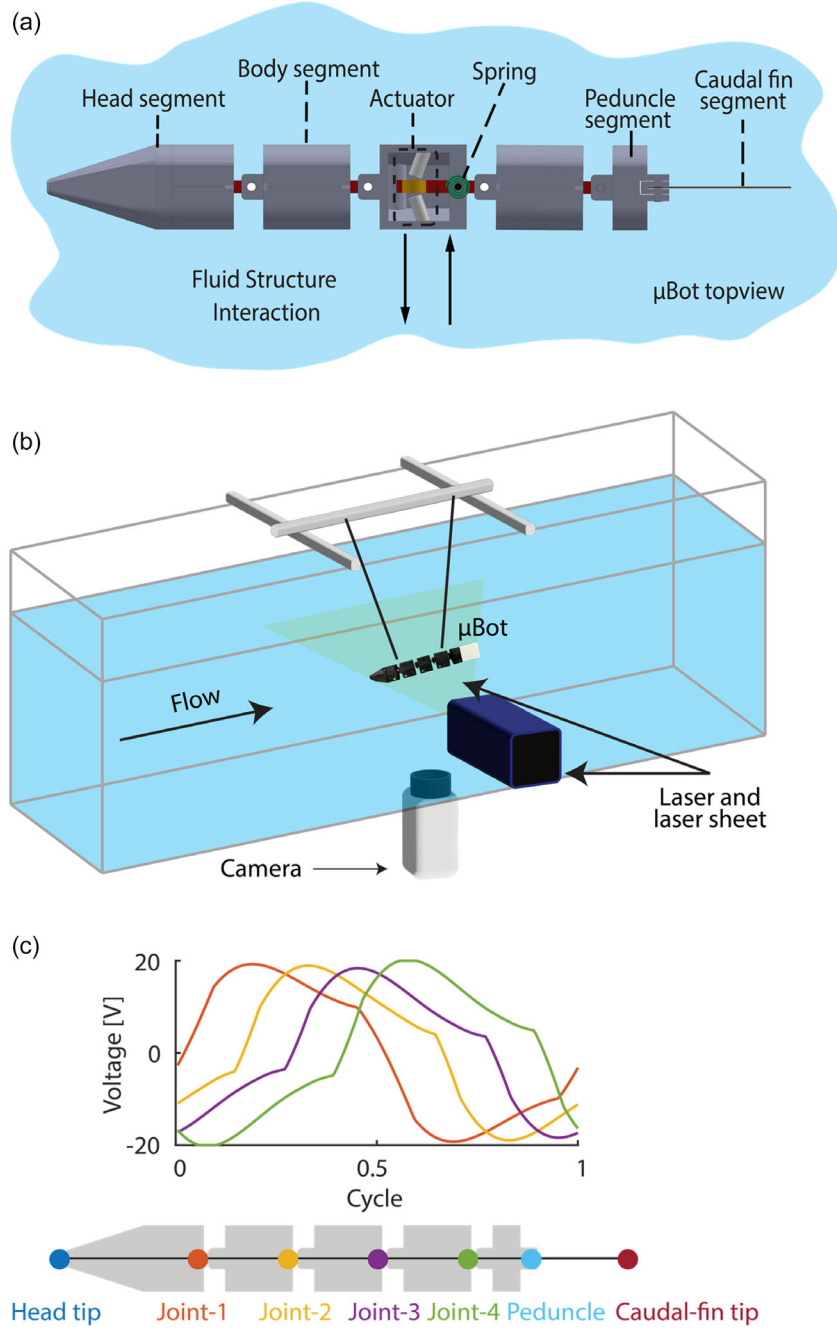


FIGURE 2 | Overview of μ Bot-4. (a) Top view of μ Bot-4 with cross section view of one body segment. There are four electromagnetic actuators located in head segment and body segments. (b) PIV experimental setup. μ Bot-4, hung in water with two thin wires, swims against the flow in a 2D plane lighted up by the laser from one side. (c) Motor control signal used in this work. Dot and curve with the same color denote the position and voltage for the same actuator. Only the frequency of control signal for SW “Res” Gaits and TW “FL” Gaits is changed.

$$\mathbf{F}_{\text{react},j} = \begin{bmatrix} \mathbf{f}_{\text{add},j} \\ \mathbf{c}_{\text{add},j} \end{bmatrix} + \begin{bmatrix} \mathbf{f}_{\text{pre},j} \\ 0 \end{bmatrix} = \mathbf{C}_a \begin{bmatrix} -\bar{m}_j l_j \omega_j v_j + \frac{1}{2} \bar{m}_j l_j^2 \omega_j^2 + C_p \cdot \frac{1}{2} \bar{m}_j (v_j^2 - v_j'^2) \\ -\bar{m}_j l_j a_j - \frac{1}{2} \bar{m}_j l_j^2 \dot{\omega}_j \\ 0 \\ 0 \\ 0 \\ -\left[\frac{1}{2} \bar{m}_j l_j^2 (a_j - \omega_j u_j) + \frac{1}{3} \bar{m}_j l_j^3 \dot{\omega}_j + \bar{m}_j l_j u_j v_j' \right] \end{bmatrix} \quad (4)$$

2.4 | Resistive Force Model

In addition to the reactive force of inviscid origin, hydrodynamic force also includes resistive force, originated from the viscous stress in the boundary layer [42]. We follow the assumptions used in Taylor’s work [66], and develop the resistive force $\mathbf{F}_{\text{resis}}$ in R^6 for j -th segment taking the form of lateral drag force and longitudinal friction force:

$$\mathbf{F}_{\text{resis},j} = -\frac{1}{2}\rho \begin{bmatrix} C_f \cdot PE_j \cdot l_j \cdot |u_j|u_j \\ C_d h_j \int_0^{l_j} |v_j|v_j dx \\ 0 \\ 0 \\ 0 \\ C_d h_j \int_0^{l_j} |v_j|v_j \cdot x dx \end{bmatrix} \quad (5)$$

where PE_j is segment perimeter and C_f, C_d are friction and drag coefficients.

2.5 | Redundancy in Hydrodynamic Parameters

Based on the reactive force model and resistive force model in Section 2.3 and 2.4, the friction force ($-\frac{1}{2}\rho \cdot C_f \cdot PE_j \cdot l_j \cdot |u_j|u_j$) and pressure force ($\frac{1}{2} \cdot C_d C_p \cdot \bar{m}(x) \cdot v_j^2(x)$) both only apply along the longitudinal direction of segments and have no component along normal direction (shown in Figure S1b). This indicates that these two forces primarily influence swimming speed by opposing each other, while having a negligible effect on swimming gaits, as they do not contribute to the normal force component. Therefore, there may exist infinitely many combinations of the friction and pressure coefficients that yield identical predictions of μBot 's swimming speed. As will be shown in the results, this potential redundancy issue can be resolved based on the PIV results on the flow and pressure fields in the wake of μBot -4.

2.6 | Body Mechanical Design Parameters

The model parameters of μBot -4 also include the mechanical design parameters of body segments (mass, inertial, and spring stiffness). The mass of each component of segment B_j is directly measured, and the moment of inertia is calculated based on the SolidWorks model (shown in Table S1). As the rubber suit also works as linear torsion spring, the spring constant is measured previously [68]. Furthermore, to improve the accuracy of spring constant estimation, correction coefficients C_s for the spring constant are applied individually for all springs and fitted in simulation.

2.7 | Simulation of μBot Swimming

The simulation was based on a simulation setup developed in authors' previous work [44]. It is composed of three parts: Gazebo with Open Dynamic Engine (ODE), Robot Operating System (ROS), and a system identification script (shown in Figure S2). The Gazebo, integrated with a hydrodynamics plugin and an actuator plugin, were used as the robot simulator. At a given timestep t_k , ODE generated actuation torque $c_{act,j}$ based on voltage $Vol_j(t_k)$ following Equation (1), calculated hydrodynamic force ($\mathbf{F}_{\text{react}}$ and $\mathbf{F}_{\text{resis}}$) based on kinematic information from t_{k-1} , and solved the equation of motion for μBot by simulating rigid body dynamics. After the μBot achieved steady forward speed, kinematic information and the forward swimming speed of μBot were sent to identification script to execute GA algorithm (see Section 4). To integrate the identification script into the Gazebo and ODE simulator, we used ROS for action controlling and data passing. During the simulation, the identification script used GA to sample policy parameters, generated modeling parameters, and sent them to Gazebo via ROS nodes and services. Then, Gazebo updated the fluid environment

and robot information, calculated robot dynamics and recorded robot kinematics information.

3 | Particle Image Velocimetry (PIV) Experiment

3.1 | PIV Experimental Setup

The PIV experiment was carried out within a self-contained water channel, featuring a transparent testing area (0.8 m width $\times$ 0.6 m height $\times$ 3.7 m length, Figure 2b). This facility could support flow velocities up to 0.9 m/s and maintain turbulence intensity under 0.1%. The robotic swimmer was suspended in the water using a pair of slender wires, which restrained its vertical movement but allowed free horizontal motion. The flow velocity in the channel was carefully adjusted so that the robot swam with a neutral pitch angle, indicating that the set flow speed matched the robot's natural swimming speed (μBot would pitch up/down if it swims faster/slower than the freestream). For the recording of the swimming gaits and fluid dynamics, we employed a 2D time-resolved PIV system provided by LaVision (LaVision Inc., Ypsilanti, MI, USA). The flow was entrained with polyamide particles (20 μm diameter, 1.03 g/cm³ density) that served as seeding agents for visualization. A diode-pumped solid-state (DPSS) laser (1W, 532 nm, class 4) was used to illuminate the horizontal plane where the robot swims at. Images of the flow were taken by a Miro 310 high speed camera fitted with a 50 mm f/1.8 D Nikon lens. The image rate was tuned based on the flow speed, with 1000 Hz at 171.2 mm/s (maximum flow speed in this work) and 400 Hz at 65.5 mm/s (minimum flow speed in this work). Details of the experimental setup can be found in our previous work [70].

3.2 | Analyses of Swimming Gaits and Flow Fields

The analysis of μBot 's swimming gaits was conducted through image data extracted using DLTdv8 software, with the specific tracking of joint positions (as marked in Figure 2c) facilitated by a deep learning network [71]. The positional data thus obtained were refined using a filter in MATLAB (R2020a, MathWorks, Natick, MA, USA) and averaged across five undulation cycles to ensure consistency and reliability in the observations.

For the derivation of velocity fields from these images, DAVIS software (version 8.2, LaVision Inc.) was employed, set to a grid resolution of 2.73 mm. These velocity fields underwent further smoothing via a Gaussian filter within MATLAB to mitigate any noise or irregularities. The refined data were then averaged over five cycles of undulation, aligning with the procedure for positional data to maintain uniformity in data processing. Utilizing the smoothed velocity fields, subsequent calculations were carried out to ascertain both vorticity and pressure fields. The estimation of pressure fields was achieved through an algorithm introduced by Dabiri et al. [72], allowing for a comprehensive analysis of the fluid dynamics around μBot .

4 | System Identification Framework

As described in Section 2, μBot swimming represents a complex FSI process consisting of a multilink robot with deformable

structures (e.g., soft rubber suit and caudal-fin) and hydrodynamics of both inviscid and viscous origins. Therefore, modeling μ Bot swimming involves a large set of heterogeneous parameters that cannot be derived purely from theory and must instead be empirically estimated from data.

4.1 | Modeling Parameters for Identification and Simplification Assumptions

Based on the analytical model described in Section 2, a total of 26 parameters, which are three hydrodynamic coefficients (C_f , C_d , C_a) for each segment, pressure force coefficients (C_p) for head and caudal-fin segment, and correction coefficient for torsion spring in each joint (C_s), need to be fitted. Such large amount of parameters render the system identification process challenging. Therefore, the following simplification was applied.

First, as a multilink robot swimming at steady speed, the longitudinal velocity of each segment in every time instance only fluctuates in a small range. Thus, we assume that all the segments have same friction coefficient C_f . To resolve the redundancy issue in parameters (see Section 2.6), we assume that for the TW “FL” Gaits, the pressure force coefficient at caudal-fin is 0 according to the results from PIV (see Section 5.1). Briefly, the PIV results showed a thrust wake and reversed Von Karman vortex structure behind the μ Bot, which indicated that the thrust generation by this gait is primarily based on circulatory forces instead of pressure forces. For the TW “FL” Gaits, as the PIV results show strong pressure gradient near the tip of the μ Bot caudal-fin, indicating that the pressure coefficient is nonzero. In addition, given that the swimming speed of SW “Res” Gaits is only twice as swimming speed of TW “FL” Gaits, we assume the friction coefficients remain same in both gait. Second, based on the gait and shape, segments can be divided into four groups (shown in Figure S1e). Thus, segments within same subset share same hydrodynamic parameters. Since the body segments and peduncle are close to the cylinder shape, we simply set the drag coefficients for segments in group 2 and 3 as 2.25, according to Taylor’s work [66]. The details of segments subsets and parameter setting are shown in Table S2. For group 3 of μ Bot-4, the deviation coefficient ΔC_a are defined as the difference of C_a from group 2 ($C_{a,2}$) and only ΔC_a are identified.

With the simplification, the total modeling parameters that need to be identified are reduced from 26 to 13, and vector γ denote these modeling parameters (shown in Figure S1e). Note that the assumptions apply on the hydrodynamic parameters are validated using sensitivity analyses.

4.2 | System Identification Method

Here, we describe the system identification methods used to estimate the modeling parameters γ for μ Bot-4 swimming at two distinct gaits measured in experiments. We first applied the identical motor control signals from the experiments to the simulation of μ Bot, which generated the predicted swimming gaits and speed. The modeling parameters were then updated iteratively using a Genetic Algorithm (GA) to match the simulation-generated swimming gaits and speed with those measured experimentally. Using this method, the physics of simulated FSI was effectively updated to match that of the experiments.

The modeling parameters estimation problem is modeled as the minimization of the objective function E_g :

$$E_g = W_{vel} \cdot E_{vel} + W_{gait} \cdot E_{gait} \quad (6)$$

$$E_{vel} = \left| \frac{u^s}{u^e} - 1 \right| \quad (7)$$

$$E_{gait} = \sum_{i=1}^n \frac{w_i}{\sum_{i=1}^n w_i} \cdot \frac{\sum_e ||G_i^e(t) - G_i^s(t)||}{\sum_s ||G_i^e(t)||}, w_i = \begin{cases} \frac{1}{2}, & i=1 \\ 5, & i=n \\ 1, & i \neq 1, n \end{cases} \quad (8)$$

where E_{vel} and E_{gait} are relative error of forward speed and gait, W_{vel} , $W_{gait} = \frac{1}{3}, \frac{2}{3}$ are the corresponding weights, respectively, for calculating the total error E_g as the cost function for GA. u is μ Bot steady-state forward swimming speed, $G_i^e(t)$ and $G_i^s(t)$ are gait functions (defined in supplementary equations SE(1)), superscripts e and s denote variables in experiments and simulations, and w_i are the corresponding weights for i -th marker on μ Bot. The previous results showed that the swimming speed is sensitive to caudal-fin motion while the gait of head tip may have negligible effect [44], thus we chose the weights accordingly. Due to the highly nonlinear and nonconvex nature of the problem, we used a GA-based approach to estimate the hydrodynamic parameters. With the motor control from experiments, GA was implemented to explore the selected modeling parameters that minimize the objective function (Equation (6)).

For each gait, the GA-based identification was performed three times with randomized initial parameter vectors to reduce sensitivity to initialization and mitigate entrapment in local optima. For SW “Res” Gait, 40 generations with a population size of 100 was performed while for TW “FL” Gait, 40 generations with a population size of 160 was performed. In each generation, the four highest-fitness individuals (fitness was defined as the reciprocal of E_g in Equation (6)) were retained as an elite set and paired for crossover to generate parent candidates for the next generation. The next-generation population was then sampled from a truncated distribution centered at the parent parameters with fixed deviations, while enforcing parameter bounds (all bounds and deviations are explicitly reported in Table S3). These hyperparameters were selected to balance (i) adequate exploration of a high-dimensional parameter space and robustness to local minima (via a moderately large population and multi-run initialization) against (ii) practical computational cost for our simulation-in-the-loop identification (via a bounded number of generations and controlled sampling deviations).

4.3 | Perturbation Test and Sensitivity Analysis

In the above system identification process, we used assumptions that the swimming gaits and speed were insensitive to a subset of hydrodynamic parameters, which were kept constant to theoretically or empirically determined values. To validate these assumptions, we performed perturbation tests and sensitivities analysis on all model parameters γ^* for both cases. A vector γ^* was defined as the combination of all identified modeling parameters and fixed hydrodynamic parameters for each case. During the perturbation test, only one parameter was perturbed each time, starting from its lower bound to its upper bound at an increment equal to 0.25% of the range. The upper and lower bounds were identical to those

used in the GA. Then with a perturbed model parameter, simulation was performed to generate the swimming gait and speed of μ Bots under the same motor control signals of the unperturbed case. The gait functions G_{ij}^p with amplitude A_{ij}^p and phase Φ_{ij}^p (shown in supplementary equations SE(2)) were fitted and the average forward speed u^p was calculated, while the subscript i denotes the index of body point and subscript j denotes the index of perturbed parameters. Normalized sensitivities S for different aspects of swimming performance are defined as follows:

$$S = \begin{cases} S_j^{A,i} = \left| \frac{A_{ij}^p - A_i^e}{A_{tail}^e} \right| \\ S_j^{\Phi,i} = |\Phi_{ij}^p - \Phi_i^e| \\ S_j^u = \left| \frac{u_j^p - u^e}{u^e} \right| \end{cases} \quad (9)$$

where $S_j^{A,i}$ and $S_j^{\Phi,i}$ denote the sensitivities of gait amplitude and gait phase of i^{th} body point against the perturbation of j^{th} parameter and S_j^u denotes the sensitivity of forward swimming speed against the perturbation of j^{th} parameter.

5 | Result

5.1 | Two Distinct Swimming Gaits and Wake Structures Emerged from Motor Frequency Responses

PIV results show that there were only two distinct types of wake structures among all the tested motor frequencies, corresponding to the two distinct swimming gaits previously measured (Figure 3 and Movie S2). From low to medium frequencies (1.72 to 6.35 Hz), the μ Bot swam at Traveling Wave “Fish-Like” gait. Using results with 2.15 Hz motor control, the μ Bot’s caudal-fin shed two pairs of vortices every cycle of movement (“2P”), consistent with prior observations [20, 21, 26] (Figure 4). When the caudal-fin tip moved from leftward, a pair of vortices (labeled as “1,” blue, clockwise, and “2,” red, counter-clockwise) was generated concurrently (Figure 4g). This pair of vortices generated a flow jet pointing leftward and downstream. Upon reversal on the left side when the fin started moving rightward, this pair of vortices shed leftward, and a new pair of vortices began to form at the caudal-fin tip, denoted as “3” and “4” in Figure 4h. There was initially a noticeable linkage between vortices “2” and “4,” but they eventually detached as the fin continued moving rightward,

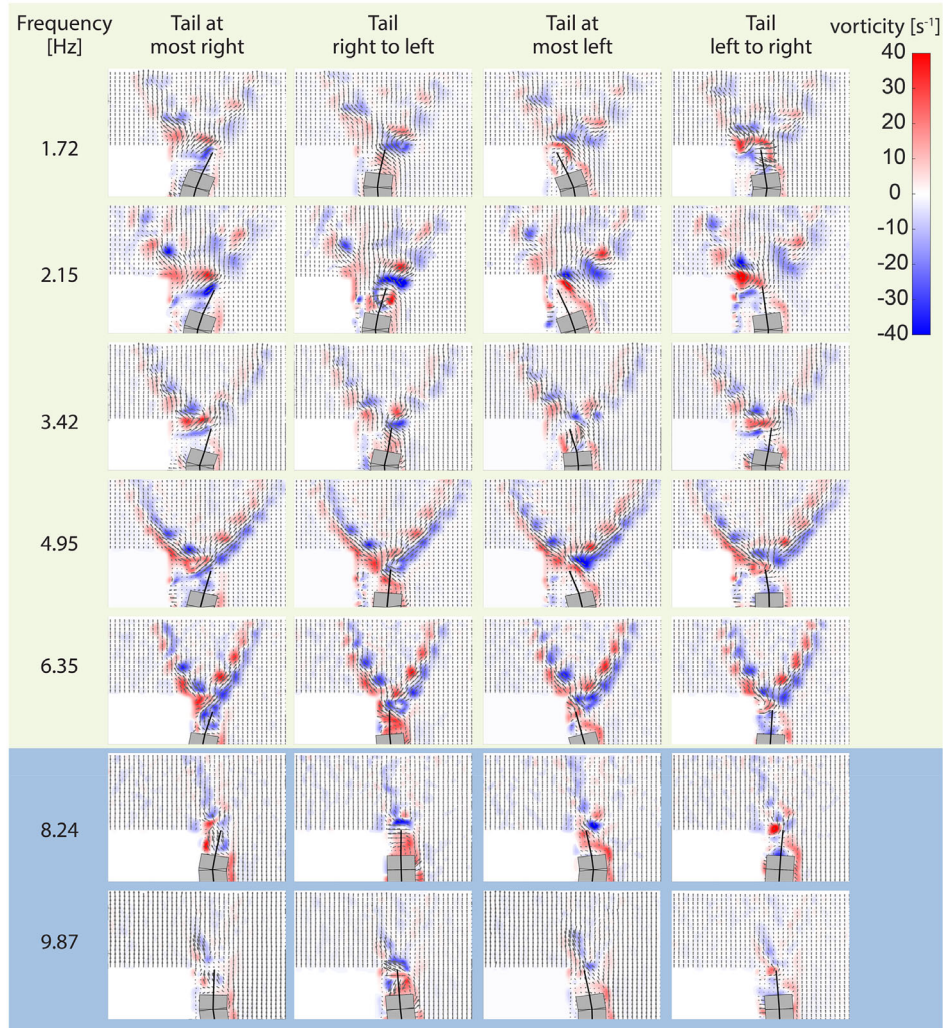


FIGURE 3 | Wake patterns of μ Bot-4 under different motor control frequency. PIV snapshots across tailbeat phases revealed that low-medium driving frequencies (1.72–6.35 Hz) produced a paired vortex shedding (“2P” wake), while high frequencies (8.24–9.87 Hz) generated a single vortices (“2S” wake).

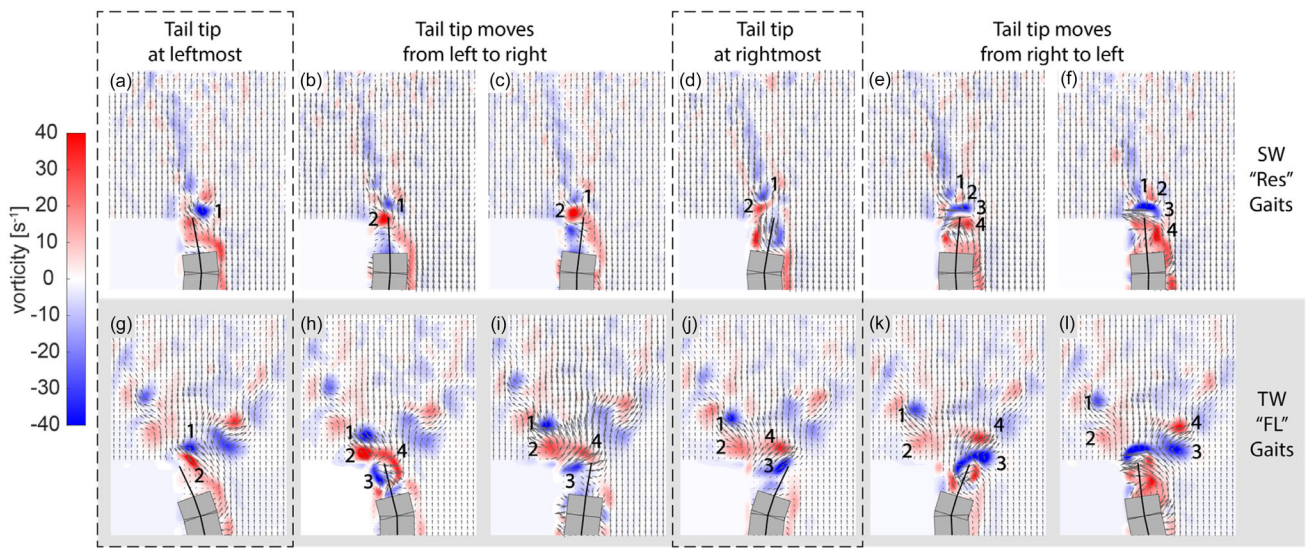


FIGURE 4 | Wake patterns of μ Bot-4 under SW “Res” Gaits and TW “FL” Gaits. Six time-instants are selected from a complete cycle of tail movement. (a–f) Wake patterns at SW “Res” Gaits and (g–l) wake patterns at TW “FL” Gaits. The tail tip is at the leftmost in (a) and (g) and moves to the rightmost in (d) to (j). The tail tip in (d)–(f)–(a) and (j)–(l)–(g) represents the returning motion from right to left. The numbers in the figures mark the vortices from the robot movement.

and the newly formed pair “3” and “4” proceeded to travel rightward together with the caudal-fin (tracing the transition from Figure 4h–j). Overall, the sequences of vortex formation for rightward movement of the fin (Figure 4j→l→g) exhibited mirrored behaviors to those of the leftward fin movement. Note that once shed from the caudal-fin, the vortices pair “1” and “2” started to dissipate and was carried downstream, as illustrated in the transition from Figure 4g–l. Collectively, these two mirrored pair of vortices (“2P”) generated a net downstream jet, together with alternating lateral jets between each pair of vortices.

At high frequencies (7.94 and 9.87 Hz), the robot swam at Standing Wave “Resonant” Gaits. Using results with 7.94 Hz motor control, the μ Bot produced a “2S” wake, shedding two single vortices per caudal-fin cycle, rather than the “2P” pattern observed under the traveling-wave fish-like gait. In the far wake, the vortex arrangement approached a von Kármán vortex street, consistent with prior reports [17–19] and characteristic of drag wakes behind blunt bodies. Putatively, it would be the same wake generated behind the μ Bot body in the absence of caudal-fin movements. On the other hand, the vortical structure near the caudal-fin exhibited stacked layers of alternating vorticities (Figure 4). Vortex “1” (blue, clockwise) was generated when the caudal-fin moved leftward (Figure 4f→a), and it shed to the left side of downstream and gradually merged with the vortices of identical polarity on the left side of the Von Karman street in the far wake (Figure 4b→c). Concurrently, vortex “2” was created and moved rightward with the caudal-fin motion (Figure 4c→d) until shed to the right side of Von Karman street downstream when the fin moved leftward (Figure 4e→f), and the same process repeated.

5.2 | Velocity Profiles Indicate Distinct Thrust Generation Mechanisms of the Two Swimming Gaits

The cycle averaged velocity profiles (in the freestream direction Y) at three wake locations are shown in Figure 5. It was clear that TW “FL” Gait generated a thrust wake, as flow velocity increased in the wake (momentum surfeit velocity profiles, Figure 5b),

representing a flow jet generated by the μ Bot in the downstream direction. As the robot accelerated the water in the wake and considering that the pressure force in Section 2.4 is caused by the velocity difference between both sides of AMV boundary, the momentum surfeit velocity profiles indicated that at low–medium frequency, the pressure force may be neglectable.

On the other hand, SW “Res” Gait generated a drag wake with a momentum deficit velocity profiles (Figure 5c). Since the wake had momentum deficit velocity profiles at SW “Res” Gait, it was hypothesized that μ Bot used pressure force to generate thrust. To confirm the hypothesis, we calculated the cycle averaged pressure field from the velocity field. The pressure fields for both wake structures were presented in Figure 5d,e, respectively. It was clear that at high frequencies, the caudal-fin motion generated a low-pressure region around the caudal-fin, while the downstream and surrounding of the low-pressure region had high pressure (Figure 5e). Therefore, the thrust could be generated due to the pressure difference. However, at low-to-medium frequencies, although the caudal-fin region had even lower pressure, the downstream and surroundings had low pressure as well (Figure 5d), resulting in a small pressure difference.

Taken together, these results indicated that there were two thrust generation mechanisms associated with the two wake structures: the low-to-medium frequency wake mainly used non-pressure force to generate thrust, while the SW “Res” Gait wake mainly used pressure force to generate thrust. These results were later used as the foundation for the determination of friction coefficients and tail pressure force coefficients in system identification process (Table 1).

5.3 | Gaits and Swimming Speed Match Closely Between Simulation and Experiment

While the identification process of modeling parameters was repeated three times for each case, GA successfully converged to a steady state, which can be seen from the result of the modeling parameter optimization. Table 2 showed the fitted modeling

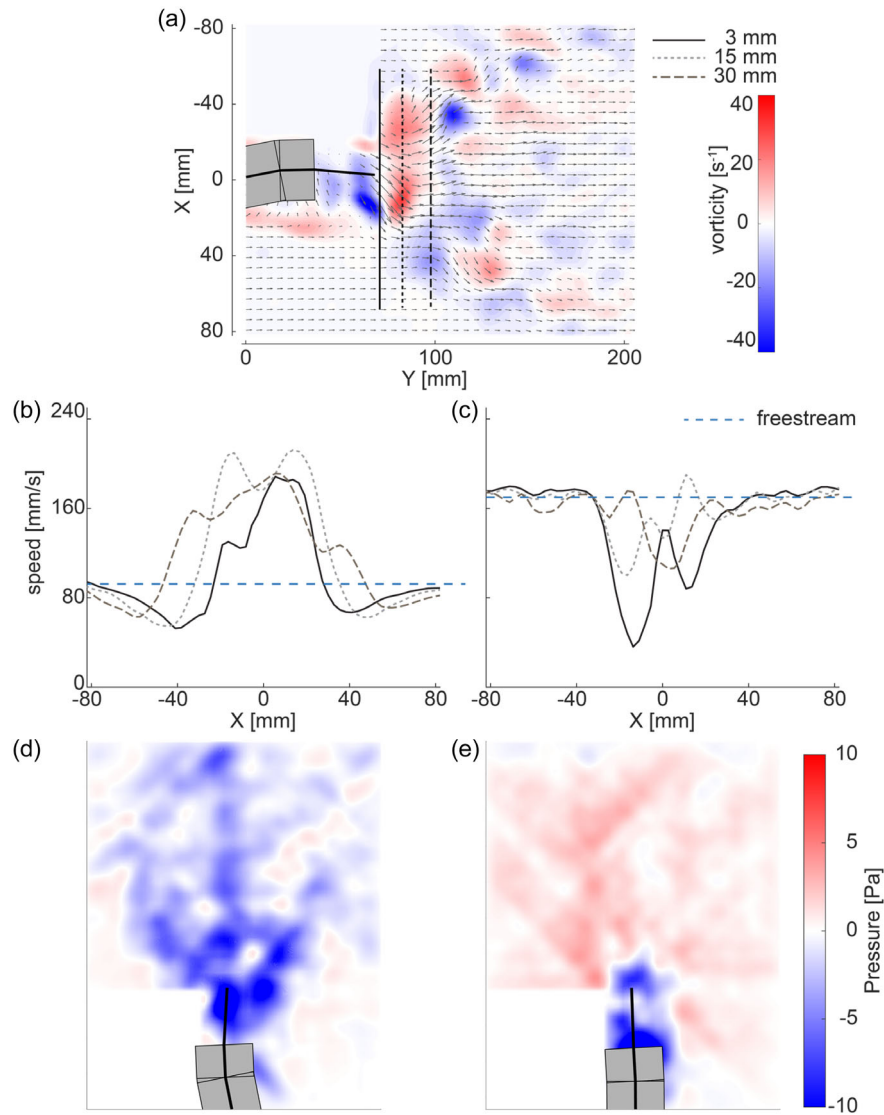


FIGURE 5 | Wake flow velocity profile and pressure fields. (a) The position of where the velocity profiles are calculated in (b) and (c) under TW “FL” Gaits. (b,c) The cycle-averaged velocity profiles in the freestream direction at the downstream of the robot caudal-fin under (b) TW “FL” Gaits and (c) SW “Res” Gaits, while the blue dotted line shows freestream speed. (d,e) Cycle-averaged pressure fields under (d) TW “FL” Gaits and (e) SW “Res” Gaits.

TABLE 1 | List of variables.

Variable	Definition
C_a, C_p, C_d, C_f	Coefficient for added mass, pressure force, drag force and friction force
C_s	Correction coefficient for spring stiffness
$G_i(t)$	Gait function for i-th body point
A_i, Φ_i	Amplitude and phase for i-th body point
u	Forward swimming speed
E_g	Objective function for system identification
E_{vel}, E_{gait}	Relative error of forward speed and gait

parameters for both cases, while Table S4 showed the fitness, relative errors of forward speed, and average relative errors of gait among all markers for each case.

As defined in Equation (6), the fitness function, which was calculated based on the relative error of forward speed E_{vel} and relative

error of gait E_{gait} , ranged from 0.370 to 0.200. With the fitted modeling parameters, our simulation can accurately predict the forward swimming speed with a relative error of less than 0.06. Compared with the error in predicted forward swimming speed, the predicted undulatory gaits showed a little bigger error of 0.296 and 0.464. Figure 1 illustrated the comparison of midline kinematics and comparison of forward speed between experiments and simulations. It can be observed clearly that not only the forward speed, but also the gait of individual segment can be predicted accurately with our system identification platform (also see Movie S1 for animated comparison between experimentally measured gaits and the simulation-predicted gaits).

5.4 | Identified Hydrodynamic Parameters Show Dependency on μ Bot Morphologies and Swimming Gaits

While the predicted swimming performance from simulation matched experimental result, the details of identified modeling

TABLE 2 | Results of identified hydrodynamic parameters and co-responding max sensitivity.

SW “Res” gaits			TW “FL” gaits		
Identified parameter values (10^{-2})		Sensitivity	Identified parameter values (10^{-2})		Theoretical values (10^{-2})
$C_{d,H}$	335	0.698	419	0.428	225 [66]
$C_{d,T}$	440	0.560	331	0.758	225 [66]
$C_{a,1}$	2.84	1.021	4.14	0.642	100 [73]
$C_{a,2}$	97.9	0.687	1.68	0.496	100 [73]
$C_{a,3}$	101	0.298	7.61	0.852	100 [73]
$C_{a,4}$	68.3	0.695	41.3	2.055	100 [73]
$C_{p,1}^*$	1.48	0.020	2.81	0.493	0 [73]
$C_{p,4}^*$	20.3	0.515	NA	1.502	100 [73]
C_f	2.48	1.544	2.48	0.744	6 [66]

parameters were shown in Table 2. The theoretical value for drag coefficients and friction coefficients were taken from Taylor’s work in 1952 [66], whereas the added mass force coefficients and pressure force coefficients were based on Lighthill’s work in 1960 [73]. For the drag coefficients of head and tails ($C_{d,H}$ and $C_{d,T}$), although the value has minor changes with gait, the result remains in a small range. For the pressure coefficient at head segment $C_{p,H}$, identified parameters for both cases were close to zero, which were similar to the assumption in Lighthill’s work [73], although the head segment is not perfectly pointed. However, the pressure at caudal-fin $C_{p,T}$ were around 0.203, less than a quarter of the estimation used in existing work [45, 61, 74, 75]. The difference may come from the assumption that velocity of fluid decreases to zero abruptly when the fluid passing boundary of AMV. For the added mass coefficients, the identified value shows different pattern among different segments. For head segment, the added mass coefficient for both cases are very close to zeros, which is similar to J. Chen et al. and J. Yu et al. [58, 75]. As the added mass coefficients remain in a small range among the body segments ($C_{a,2}$ and $C_{a,3}$), there were substantial differences, while μ Bot are under the different frequencies and gaits. With TW “FL” Gait, the identified C_a of the body segments for μ Bot-4 were close to zero and similar to the estimated results of Chen et al. and Yu et al. [57, 75]. For μ Bot-4 with SW “Res” Gait, the identified $C_{a,2}$ and $C_{a,3}$ for the body segments were much higher and closed to the theoretical value (1.0), although the sensitivity results showed that the swimming performance were not very sensitive to $C_{a,3}$. For both groups of gaits, the added mass coefficient of caudal-fin segment reminded close to the 0.5, very close to the result of Porez et al. [45]. These inconsistency across gaits indicated that the relationship between added mass coefficients and swimming gait is still unrevealed and the identified hydrodynamic parameters should not be interpreted as universally transferable parameters that can be directly generalized across substantially different Reynolds-number regimes without re-identification. For the resistive model coefficients, the drag coefficient C_d showed moderate changes between SW “Res” Gait and TW “FL” Gait.

5.5 | Swimming Performance of μ Bot Is Sensitive to a Subset of Hydrodynamic Parameters

To validate our selection of modeling parameters, perturbation tests were designed and described in Section 4.3, and the results were shown in Figure S3. In Figure S3, it was clearly shown that all the parameters that have a non-neglectable effect on swimming performance were located at the left, which were the fitted modeling parameters. Based on the results, we confirmed our assumption that in both cases, C_p and C_f had neglectable influence on undulation amplitude of all segments, but showed critical effect on swimming speed. This can simplify the system identification process by decoupling these two hydrodynamic parameter with others. However, the swimming speed of μ Bot under TW “FL” gait is not only sensitive to C_p and C_f . In the TW “FL” gait, steady forward swimming arises from a balance among drag, friction, and added-mass forces, rendering the swimming speed sensitive to most model parameters. Emerging from the complex FSI, the swimming gaits are affected by both drag force and added mass force, and our results do not show the dominated force and torque for gait of each segment.

Through the perturbation tests, we validated that the selected modeling parameters for identification were appropriate and that μ Bot gaits and swimming speeds were sensitive to these parameters.

5.6 | Actuators Functioning also Matched Closely Between Simulation and Experiment

While the diverse gaits of μ Bot emerged from different motor control frequencies, the actuator functions and work loop showed strong frequency dependency as well (Figure 6b). At SW “Res” Gait, the posterior actuators of μ Bot-4 (last 2 actuator) behaved as virtual springs (leftward inclined work loop) with low power output (small, enclosed area); together with the rubber suit elasticity, they effectively stiffened the posterior body of the robots. At TW “FL” Gait, all the actuators behaved as motor combined with virtual mass, thus softening the body of μ Bots. These showed similar results to our previous work [62].

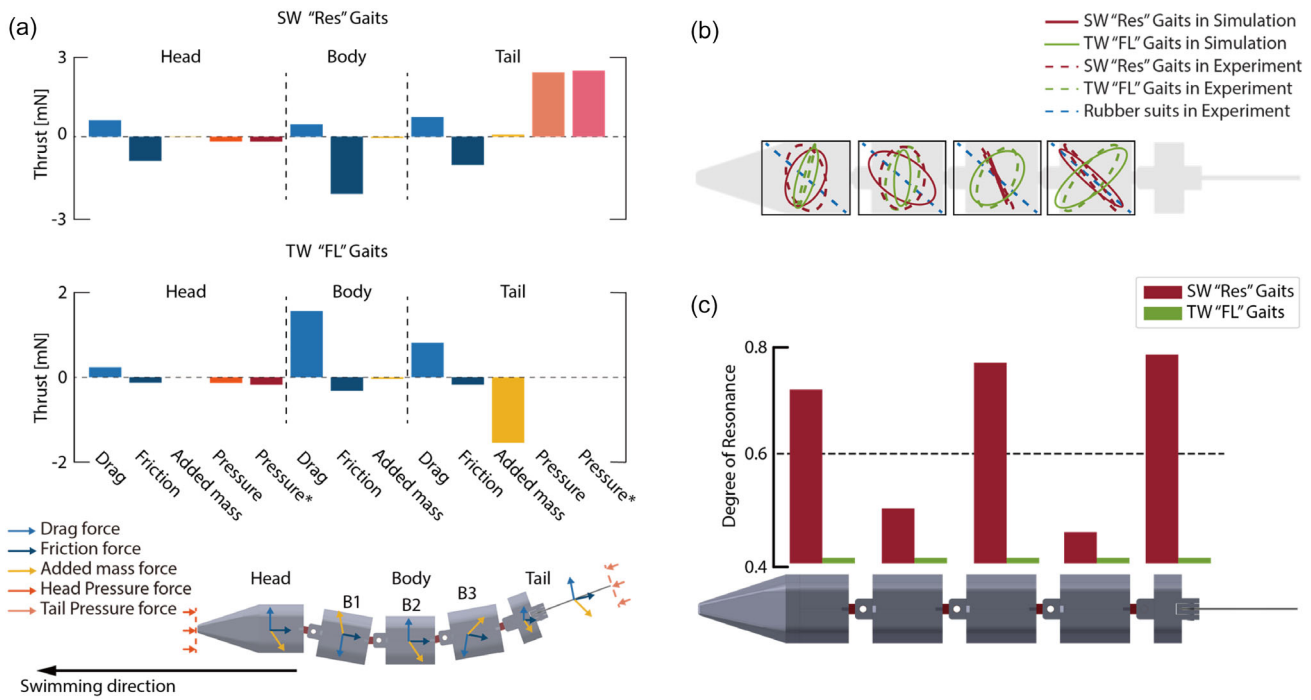


FIGURE 6 | (a) Thrust generation mechanisms of μ Bot-4 under TW “FL” Gaits and SW “Res” Gaits. The plot on the top shows the thrust of μ Bots with SW “Res” Gait and the plots in the middle show the thrust of μ Bot with TW “FL” Gait. The plot at the bottom illustrates different hydrodynamic forces on the segments. (b) Power characteristics of μ Bots in experiments and simulations at TW “FL” Gaits and SW “Res” Gaits. (c) Measurement of degree of resonance. The measurements are the remapped RMS between momentum of VM and power output of VS starting from the located segments to the caudal-fin. Higher value of the measurement denotes that the selected segments were closer to the resonance state.

5.7 | Roles of Various Hydrodynamic Mechanisms

With the identified modeling parameters, our model can be used to analyze the role of different hydrodynamic forces in thrust mechanisms and diverse gait emerge from FSI.

5.7.1 | Thrust Generation Mechanism Varies with Swimming Gaits

Figure 6a showed the average thrust over one undulation period from different hydrodynamic forces of both cases. We categorized the segments as head (head segment), body (three body segments) and tail (peduncle and caudal-fin). Our results, supporting our conclusion in Section 5.5 show that the thrust generation mechanisms of both cases are very different. In SW ‘Res’ Gait, the thrust force is mainly generated from tail pressure force and drag force, while the contribution of added mass force is negligible. In the existing research, the thrust either comes from drag force or tail pressure force, and it is rare to see where the robot use both drag force and tail pressure force to generate thrust under one gait. The resisting force for forward swimming is dominated by friction force. When it comes to TW ‘FL’ Gait, the thrust generation mechanism changes. As the tail pressure force coefficient was set to zero, the main source of thrust becomes drag force, same with the case from Chen et al. [57]. The contribution of added mass force is no longer negligible as it shows comparable amplitude with friction force. In additional to the original cases where μ Bot could swim freely in the environment, we fixed the head tip of μ Bot so that the forward swimming speed was zero but kept the gaits of μ Bot in both cases. Our results showed that the fluid pressure forces at head and tail (Pressure Head* and Pressure Tail*) in both cases with zero forward swimming speed

only had very minor difference comparing with pressure forces (Pressure Head, Pressure Tail) with the full forward speed under with SW ‘Res’ Gait and TW ‘FL’ Gaits.

Our results show that with a simple tuning of frequency of control signal, not only the swimming gait and speed change, the thrust generation mechanism changes. As the role of different hydrodynamic force in thrust generation changes with gait, this result emphasized the important role of swimming gait optimization in maximizing forward swimming speed.

5.7.2 | Resonance Observed in SW ‘Res’ Gait but Absent in TW ‘FL’ Gait

μ Bot models not only contained the rigid body of segments but also drag force, added mass force and conservative torque from rubber suits. The drag force has negative quadratic relation with segmental lateral velocity, which behaves as damping force. Added mass force has linear relation with segmental acceleration which behaves like additional mass attached to segments. Thus, μ Bot model can be regarded as the mass-spring-damper system. The virtual damping force (VD) was the drag force, the virtual mass (VM) was composed of μ Bot segments and added mass within AMV around the segment, and the virtual spring (VS) contained both lumped rubber suits and actuators.

While the system reaches the resonance state of the mass-spring-damper system, the kinetic energy and potential energy have the same fluctuation amplitude but a phase delay of π , which means the kinetic system only transfers energy with springs and vice versa. In that scenario, RMS between the momentum of VM and the power output of VS zero out while reaching resonance state and the smaller RMS is, the closer to resonance.

Thus, we designed a dimensionless number to measure the degree of resonance (DoR) within diverse gaits. We calculated the RMS between the momentum of VM and the power output of VS starting from different segments to the tail in both cases and mapped it from $[0, \infty]$ to $[0, 1]$, while 0 means no resonance, and 1 means resonance. The measurement of resonance K_r was defined as:

$$K_{r,i} = e^{-\frac{R}{M}S_i^4} \quad (10)$$

$$\text{RMS}_i = \sqrt{\frac{\sum_{t=0}^T (P_{VM,i,t} - P_{VS,i,t})^2}{\sum_{t=0}^T (P_{VM,i,t})^2}}, i = 1, 2, \dots, 5 \quad (11)$$

where P_{VM} and P_{VS} denote instantaneous momentum of VM and the power output of VS, subscript i denotes the power/momentum summed from i -th segments to tail, and subscript t denotes the timestep which ranged from zeros to one undulation period. According to the definition, the higher K_r indicates that the gait is closer to resonance state.

From Figure 6c, it is clearly shown that with SW “Res” Gait, $K_{r,1}, K_{r,3}, K_{r,5}$ in $\mu\text{Bot-4}$ were significantly higher than the rest of K_r , which indicated that in the SW “Res” Gait case of $\mu\text{Bot-4}$, segments can be regarded as three groups: (1) head segment, (2) first body segment and second body segment, and (3) third body segment, peduncle, and caudal-fin, and each group reached the resonance state. These results may indicate that the SW “Res” Gaits resulted from the state near resonance of fluid–structure interaction. However, in TW “FL” Gait, measurements of resonance K_r were small, which suggest that there was no resonance under TW “FL” Gait.

6 | Conclusion and Future Work

In this work, we uncovered the thrust generation mechanisms underlying two distinct swimming gaits of a modular robotic swimmer (μBot). This was achieved via combining PIV measurements, an improved Lighthill analytical model with partially unknown parameters, and system identification methods. Frequency responses of μBot showed that its swimming gaits transitioned from TW “FL” gaits producing a thrust wake at low-to-media frequency to SW “Res” gaits producing a drag wake at high frequency. Our results showed that these two gaits relied on fundamentally distinct thrust generation mechanisms. With TW “FL” gaits, backward-propagating waves at low frequency propelled flow downstream and generated a jet in the wake (i.e., a thrust wake with “2P” pattern), while the thrust arose from circulatory drag forces that was captured by the resistive model. In contrast, with the SW “Res” gait, rapid caudal-fin undulation created a low-pressure region surrounding the caudal fin (Figure 5e), leading to a high-pressure gradient between the tip region of the caudal fin and the downstream wake. The downstream wake however resembles a drag wake exhibiting reversed von Karman vortex with “2S” pattern. This corresponds to a pressure-based thrust generating mechanism that is consistent with classical predictions of Lighthill’s model [73]. Resonance analysis further demonstrated that only SW “Res” gait exhibited resonance-like energy exchange.

The application of PIV measurements was essential to resolve the parameter redundancies in the system identification of the improved Lighthill analytical model, enabling more reliable

estimation of hydrodynamic forces. Analyses of hydrodynamic forces in simulation highlighted the complexity of fluid–structure interactions, demonstrating that the changes in motor frequency can fundamentally alter the hydrodynamic parameters in the model and therefore the thrust production mechanisms. While this study focused on a single robotic swimmer under two gaits, we believe that the framework is broadly applicable to robots with diverse morphologies and control strategies. Going forward, extending this integrative approach to more complex motion primitives, flexible structures, and collective swimming behaviors will deepen our understanding of gait-dependent propulsion and accelerate the design of efficient, adaptive bio-inspired robots for exploration, monitoring, and marine technologies.

Author Contributions

Donghao Li: conceptualization (lead), data curation (lead), formal analysis (lead), investigation (lead), methodology (lead), software (lead), validation (lead), visualization (lead), writing – original draft (lead), and writing – review and editing (lead). **Hankun Deng:** conceptualization (supporting), data curation (supporting), formal analysis (equal), investigation (supporting), methodology (supporting), software (supporting), visualization (equal), writing – original draft (equal), and writing – review and editing (supporting). **Kundan Panta:** conceptualization (supporting), formal analysis (supporting), investigation (supporting), methodology (supporting), and writing – original draft (supporting). **Azar Panah:** conceptualization (supporting), formal analysis (supporting), funding acquisition (lead), investigation (supporting), methodology (supporting), project administration (supporting), supervision (supporting), and writing – original draft (supporting). **Bo Cheng:** formal analysis (supporting), funding acquisition (lead), investigation (lead), methodology (equal), project administration (lead), resources (lead), software (supporting), supervision (lead), writing – original draft (equal), and writing – review and editing (equal).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data and code are available on <https://github.com/DonghaoLi-PSU/MUBot---System-identification-and-PIV>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.